CHAPTER C2
COMBINATIONS OF LOADS

C2.1 GENERAL

Loads in this standard are intended for use with design specifications for conventional structural materials, including steel, concrete, masonry, and timber. Some of these specifications are based on allowable stress design, whereas others use strength (or limit states) design. In the case of allowable stress design, design specifications define allowable stresses that may not be exceeded by load effects caused by unfactored loads, that is, allowable stresses contain a factor of safety. In strength design, design specifications provide load factors and, in some instances, resistance factors. Load factors given herein were developed using a first-order probabilistic analysis and a broad survey of the reliabilities inherent in contemporary design practice (Ellingwood et al. 1982; Galambos et al. 1982). It is intended that these load factors be used by all material-based design specifications that adopt a strength design philosophy in conjunction with nominal resistances and resistance factors developed by individual materials-specification-writing groups. Ellingwood et al. (1982) also provide guidelines for materials-specification-writing groups to aid them in developing resistance factors that are compatible, in terms of inherent reliability, with load factors and statistical information specific to each structural material.

The requirement to use either allowable stress design (ASD) or load and resistance factor design (LRFD) dates back to the introduction of load combinations for strength design (LRFD) in the 1982 edition of the standard. An indiscriminate mix of the LRFD and ASD methods may lead to unpredictable structural system performance because the reliability analyses and code calibrations leading to the LRFD load combinations were based on member rather than system limit states. Registered design professionals often design (or specify) cold-formed steel and open web steel joists using ASD and, at the same time, design the structural steel in the rest of the building or other structure using LRFD. Foundations are also commonly designed using ASD, although strength design is used for the remainder of the structure. Using different design standards for these types of elements has not been shown to be a problem. This requirement is intended to permit current industry practice while, at the same time, not permitting LRFD and ASD to be mixed indiscriminately in the design of a structural frame.

C2.2 SYMBOLS

Self-straining forces and effects can be caused by differential settlement of foundations, creep, shrinkage or expansion in concrete members after placement and similar effects that depend on the material of construction and conditions of constraint, and changes in temperature of members caused by environmental conditions or operational activities during the service life of the structure. See Section C1.3.3 for examples of when self-straining forces and effects may develop.

C2.3 LOAD COMBINATIONS FOR STRENGTH DESIGN

C2.3.1 Basic Combinations. Unfactored loads to be used with these load factors are the nominal loads of this standard. Load factors are from Ellingwood et al. (1982), with the exception of the load factor of 1.0 for E, based on the more recent NEHRP research on seismic-resistant design (FEMA 2004), and for W, based on the wind speed maps at longer return periods for each Risk Category. The basic idea of the load combination analysis is that in addition to dead load, which is considered to be permanent, one of the principal loads (previously referred to as primary variable loads) takes on its maximum lifetime value while the other loads assume "arbitrary point-in-time" values, the latter being loads that would be measured at any instant of time (Turkstra and Madsen 1980). This is consistent with the manner in which loads actually combine in situations in which strength limit states may be approached. However, nominal loads in this standard are substantially in excess of the arbitrary point-in-time values. To avoid having to specify both a maximum and an arbitrary point-in-time value for each load type, some of the specified load factors are less than unity in combinations 2 through 5. Load factors in Section 2.3.1 are based on a survey of reliabilities inherent in existing design practice (Ellingwood et al. 1982; Galambos et al. 1982).

In design where first-order analysis is permitted, superposition of factored loads can be performed either before or after the analysis. However, when a second-order analysis, which considers the effects of structural deformation on member forces, is used to design members and connections, the load factors must be applied before the analysis. The second-order analysis can be accomplished using a computer program with this capability for frame effects and member effects or by amplifying the results of a first-order analysis through the use of coefficients that amplify the first-order moments for the effects of member deformations or joint displacements. Since second-order effects are nonlinear, the second-order analyses must be conducted under factored load combinations (strength design) or load combinations amplified by the factor of safety (allowable stress design). Second-order effects in this context are the effects of loads acting on the deformed configuration of a structure and include P–δ effects and P–Δ effects.

Note that each of the principal loads, including the dead load, is a random variable. The degree of variability in each load is reflected in the associated load factor for the principal load; the other load factors provide the companion values.

The principal loads in the load combinations are identified in Table C2.3-1. The load factor on wind loads in combinations

Table C2.3-1 Principal Loads for Strength Design Load Combinations

Load Combination		Principal Load
1	$1.4D$	D
2	$1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$	L
3	$1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (1.0L \text{ or } 0.5W)$	L_r or S or R
4	$1.2D + 1.0W + 1.0L + 0.5(L_r \text{ or } S \text{ or } R)$	W
5	$0.9D + 1.0W$	W
6	$1.2D + E_v + E_h + L + 0.2S$	E
7	$0.9D - E_v + E_h$	E

4 and 5 of Section 2.3.1 and on earthquake loads in combinations 6 and 7 of Section 2.3.6 were changed to 1.0 in previous editions of ASCE 7 when the maps for design wind speed and design seismic acceleration were modified to support a risk-consistent design methodology, as described in Chapters 21 and 26.

Exception 2 permits the companion load S that appears in combinations 2 and 4 to be the balanced snow load defined in Sections 7.3 for flat roofs and 7.4 for sloped roofs. Drifting and unbalanced snow loads, as principal loads, are covered by combination 3.

Load combinations 5 and 7 apply specifically to the case in which the structural actions due to lateral forces and gravity loads counteract one another.

Load combination requirements in Section 2.3 apply only to strength limit states. Serviceability limit states and associated load factors are covered in Appendix C of this standard.

This standard historically has provided specific procedures for determining magnitudes of dead, occupancy live, wind, snow, and earthquake loads. Other loads not traditionally considered by this standard may also require consideration in design. Some of these loads may be important in certain material specifications and are included in the load criteria to enable uniformity to be achieved in the load criteria for different materials. However, statistical data on these loads are limited or nonexistent, and the same procedures used to obtain load factors and load combinations in Section 2.3.1 cannot be applied at the present time. Accordingly, load factors for fluid load (F) and lateral pressure caused by soil and water in soil (H) have been chosen to yield designs that would be similar to those obtained with existing specifications, if appropriate adjustments consistent with the load combinations in Section 2.3.1 were made to the resistance factors. Further research is needed to develop more accurate load factors.

Fluid load, F, defines structural actions in structural supports, framework, or foundations of a storage tank, vessel, or similar container caused by stored liquid products. The product in a storage tank shares characteristics of both dead and live loads. It is similar to a dead load in that its weight has a maximum calculated value, and the magnitude of the actual load may have a relatively small dispersion. However, it is not permanent: Emptying and filling cause fluctuating forces in the structure, the maximum load may be exceeded by overfilling, and densities of stored products in a specific tank may vary.

Uncertainties in lateral forces from bulk materials, included in H, are higher than those in fluids, particularly when dynamic effects are introduced as the bulk material is set in motion by filling or emptying operations. Accordingly, the load factor for such loads is set equal to 1.6.

Where H acts as a resistance, a factor of 0.9 is suggested if the passive resistance is computed with a conservative bias. The intent is that soil resistance be computed for a deformation limit appropriate for the structure being designed, not at the ultimate passive resistance. Thus an at-rest lateral pressure, as defined in the technical literature, would be conservative enough. Higher resistances than at-rest lateral pressure are possible, given appropriate soil conditions. Fully passive resistance would likely not ever be appropriate because the deformations necessary in the soil would likely be so large that the structure would be compromised. Furthermore, there is a great uncertainty in the nominal value of passive resistance, which would also argue for a lower factor on H should a conservative bias not be included.

C2.3.2 Load Combinations Including Flood Load. The nominal flood load, F_a, is based on the 100-year flood (Section 5.1). The recommended flood load factor of 2.0 in V-Zones and Coastal A-Zones is based on a statistical analysis of flood loads associated with hydrostatic pressures, pressures caused by steady overland flow, and hydrodynamic pressures caused by waves, as specified in Section 5.4.

The flood load criteria were derived from an analysis of hurricane-generated storm tides produced along the United States East and Gulf coasts (Mehta et al. 1998), where storm tide is defined as the water level above mean sea level resulting from wind-generated storm surge added to randomly phased astronomical tides. Hurricane wind speeds and storm tides were simulated at 11 coastal sites based on historical storm climatology and on accepted wind speed and storm surge models. The resulting wind speed and storm tide data were then used to define probability distributions of wind loads and flood loads using wind and flood load equations specified in Sections 5.3 and 5.4. Load factors for these loads were then obtained using established reliability methods (Ellingwood et al. 1982; Galambos et al. 1982) and achieve approximately the same level of reliability as do combinations involving wind loads acting without floods. The relatively high flood load factor stems from the high variability in floods relative to other environmental loads. The presence of $2.0F_a$ in both combinations (4) and (6) in V-Zones and Coastal A-Zones is the result of high stochastic dependence between extreme wind and flood in hurricane-prone coastal zones. The $2.0F_a$ also applies in coastal areas subject to northeasters, extratropical storms, or coastal storms other than hurricanes, where a high correlation exists between extreme wind and flood.

Flood loads are unique in that they are initiated only after the water level exceeds the local ground elevation. As a result, the statistical characteristics of flood loads vary with ground elevation. The load factor 2.0 is based on calculations (including hydrostatic, steady flow, and wave forces) with stillwater flood depths ranging from approximately 4 to 9 ft (1.2–2.7 m) (average stillwater flood depth of approximately 6 ft (1.8 m)) and applies to a wide variety of flood conditions. For lesser flood depths, load factors exceed 2.0 because of the wide dispersion in flood loads relative to the nominal flood load. As an example, load factors appropriate to water depths slightly less than 4 ft (1.2 m) equal 2.8 (Mehta et al. 1998). However, in such circumstances, the flood load generally is small. Thus, the load factor 2.0 is based on the recognition that flood loads of most importance to structural design occur in situations where the depth of flooding is greatest.

C2.3.3 Load Combinations Including Atmospheric Ice Loads. Load combinations 2, 4, and 5 in Section 2.3.3 and load combinations 2, 3, and 7 in Section 2.4.3 include the simultaneous effects of snow loads as defined in Chapter 7 and atmospheric ice loads as defined in Chapter 10. Load combinations 2 and 3 in Sections 2.3.3 and 2.4.3 introduce the simultaneous effect of wind on the atmospheric ice. The ice load, D_i, and the wind load on the atmospheric ice, W_i, in combination correspond to an event with approximately a 500-year mean recurrence interval (MRI). Accordingly, the load factors on W_i and D_i are set equal to 1.0 and 0.7 in Sections 2.3.3 and 2.4.3, respectively. The 0.7 load factor on W_i and D_i in Section 2.4.3 aligns allowable stress design to have reliabilities for atmospheric ice loads consistent with the definition of wind and ice loads in Chapter 10 of this standard, which is based on strength principles. The snow loads defined in Chapter 7 are based on measurements of frozen precipitation accumulated on the ground, which includes snow, ice caused by freezing rain, and rain that falls onto snow and later freezes. Thus the effects of freezing rain are included in the snow loads for roofs, catwalks, and other surfaces to which snow loads are normally applied. The atmospheric ice loads defined in Chapter 10 are applied simultaneously to those portions of the structure on which ice caused by freezing rain, in-cloud icing, or snow accrete that are not subject to the snow loads in Chapter 7. A trussed tower installed on the roof of a building is one example. The snow loads from Chapter 7 would be applied to the roof with the atmospheric ice loads from Chapter 10 applied to the trussed tower. Section 2.3.3 load combination 2 ($1.2D + L + D_i + W_i + 0.5S$) or Section 2.4.3 load combination 2 ($D + 0.7D_i + 0.7W_i + S$) are applicable. If a trussed tower has working platforms, the snow loads would be applied to the surface of the platforms, similar to a roof, with the atmospheric ice loads applied to the tower. Section 2.3.3 load combination 2 would reduce to $1.2D + D_i$ for cases where the live load, wind on ice load, and snow load are zero. Section 2.4.3 load combination 2 would similarly reduce to $D + 0.7D_i$. If a sign is mounted on a roof, the snow loads would be applied to the roof and the atmospheric ice loads to the sign.

C2.3.4 Load Combinations Including Self-Straining Forces and Effects. Self-straining forces and effects should be calculated based on a realistic assessment of the most probable values rather than the upper bound values of the variables. The most probable value is the value that can be expected at any arbitrary point in time.

When self-straining forces and effects are combined with dead loads as the principal action, a load factor of 1.2 may be used. However, when more than one variable load is considered and self-straining forces and effects are considered as a companion load, the load factor may be reduced if it is unlikely that the principal and companion loads will attain their maximum values at the same time. The load factor applied to T should not be taken as less than a value of 1.0.

If only limited data are available to define the magnitude and frequency distribution of the self-straining forces and effects, then its value must be estimated conservatively. Estimating the uncertainty in the self-straining forces and effects may be complicated by variation of the material stiffness of the member or structure under consideration.

When checking the capacity of a structure or structural element to withstand the effects of self-straining forces and effects, the following load combinations should be considered.

When using strength design:

$$1.2D + 1.2T + 0.5L$$

$$1.2D + 1.6L + 1.0T$$

These combinations are not all-inclusive, and judgment is necessary in some situations. For example, where roof live loads or snow loads are significant and could conceivably occur simultaneously with self-straining forces and effects, their effect should be included. The design should be based on the load combination causing the most unfavorable effect.

C2.3.5 Load Combinations for Nonspecified Loads. Engineers may wish to develop load criteria for strength design that are consistent with the requirements in this standard in some situations where the standard provides no information on loads or load combinations. They also may wish to consider loading criteria for special situations, as required by the client in performance-based engineering (PBE) applications in accordance with Section 1.3.1.3. Groups responsible for strength design criteria for design of structural systems and elements may wish to develop resistance factors that are consistent with the standard. Such load criteria should be developed using a standardized procedure to ensure that the resulting factored design loads and load combinations will lead to target reliabilities (or levels of performance) that can be benchmarked against the common load criteria in Section 2.3.1. Section 2.3.5 permits load combinations for strength design to be developed through a standardized method that is consistent with the methodology used to develop the basic combinations that appear in Section 2.3.1.

The load combination requirements in Section 2.3.1 and the resistance criteria for structural steel in AISC 360 (2016), for cold-formed steel in AISI S100 (2016), for structural concrete in ACI 318 (2014), for structural aluminum in the *Specification for Aluminum Structures* (Aluminum Association 2015), for engineered wood construction in *AWC NDS-2015 National Design Specification for Wood Construction* (2015), and for masonry in TMS 402-16, *Building Code Requirements and Specifications for Masonry Structures and Companion Commentaries* (2016), are based on modern concepts of structural reliability theory. In probability-based limit states design (PBLSD), the reliability is measured by a reliability index, β, which is related (approximately) to the limit state probability by $P_f = \Phi(-\beta)$. The approach taken in PBLSD was to

1. Determine a set of reliability objectives or benchmarks, expressed in terms of β, for a spectrum of traditional structural member designs involving steel, reinforced concrete, engineered wood, and masonry. Gravity load situations were emphasized in this calibration exercise, but wind and earthquake loads were considered as well. A group of experts from material specifications participated in assessing the results of this calibration and selecting target reliabilities. The reliability benchmarks so identified are *not* the same for all limit states; if the failure mode is relatively ductile and consequences are not serious, β tends to be in the range 2.5 to 3.0, whereas if the failure mode is brittle and consequences are severe, β is 4.0 or more.
2. Determine a set of load and resistance factors that best meets the reliability objectives identified in (1) in an overall sense, considering the scope of structures that might be designed by this standard and the material specifications and codes that reference it.

The load combination requirements appearing in Section 2.3.1 used this approach. They are based on a "principal action–companion action" format, in which one load is taken at its maximum value while other loads are taken at their point-in-time values. Based on the comprehensive reliability analysis performed to support their development, it was found that these load factors are well approximated by

$$\gamma_Q = (\mu_Q/Q_n)(1 + \alpha_Q \beta V_Q) \qquad \text{(C2.3-1)}$$

in which μ_Q is the mean load, Q_n is the nominal load from other chapters in this standard, V_Q is the coefficient of variation in the load, β is the reliability index, and α_Q is a sensitivity coefficient that is approximately equal to 0.8 when Q is a principal action and 0.4 when Q is a companion action. This approximation is valid for a broad range of common probability distributions used to model structural loads. The load factor is an increasing function of the bias in the estimation of the nominal load, the variability in the load, and the target reliability index, as common sense would dictate.

As an example, the load factors in combination 2 of Section 2.3.1 are based on achieving a β of approximately 3.0 for a ductile limit state with moderate consequences (e.g., formation of first plastic hinge in a steel beam). For live load acting as a principal action, $\mu_Q/Q_n = 1.0$ and $V_Q = 0.25$; for live load acting as a companion action, $\mu_Q/Q_n \approx 0.3$ and $V_Q \approx 0.6$. Substituting these statistics into Eq. (C2.3-1), $\gamma_Q = 1.0[1 + 0.8(3)(0.25)] = 1.6$ (principal action) and $\gamma_Q = 0.3[1 + 0.4(3)(0.60)] = 0.52$ (companion action). ASCE Standard 7-05 (2005) stipulates 1.60 and 0.50 for these live load factors in combinations 2 and 3. If an engineer wished to design for a limit state probability that is less than the standard case by a factor of approximately 10, β would increase to approximately 3.7, and the principal live load factor would increase to approximately 1.74.

Similarly, resistance factors that are consistent with the aforementioned load factors are well approximated for most materials by

$$\phi = (\mu_R/R_n)\exp[-\alpha_R \beta V_R] \qquad \text{(C2.3-2)}$$

in which μ_R = mean strength, R_n = code-specified strength, V_R = coefficient of variation in strength, and α_R = sensitivity coefficient, equal approximately to 0.7. For the limit state of yielding in an ASTM A992 (2011) steel tension member with specified yield strength of 50 ksi (345 MPa), $\mu_R/R_n = 1.06$ (under a static rate of load) and $V_R = 0.09$. Eq. (C2.3-2) then yields $\phi = 1.06\exp[-(0.7)(3.0)(0.09)] = 0.88$. The resistance factor for yielding in tension in Section D of the *AISC Specification* (2010) is 0.9. If a different performance objective were to require that the target limit state probability be decreased by a factor of 10, then ϕ would decrease to 0.84, a reduction of about 7%. Engineers wishing to compute alternative resistance factors for engineered wood products and other structural components where duration-of-load effects might be significant are advised to review the reference materials provided by their professional associations before using Eq. (C2.3-2).

There are two key issues that must be addressed to use Eqs. (C2.3-1) and (C2.3-2): selection of reliability index, β, and determination of the load and resistance statistics.

The reliability index controls the safety level, and its selection should depend on the mode and consequences of failure. The loads and load factors in this standard do not explicitly account for higher reliability indices normally desired for brittle failure mechanisms or more serious consequences of failure. Common standards for design of structural materials often do account for such differences in their resistance factors (for example, the design of connections under AISC or the design of columns under ACI). Tables 1.3-1 and 1.3-2 provide general guidelines for selecting target reliabilities consistent with the extensive calibration studies performed earlier to develop the load requirements in Section 2.3.1 and the resistance factors in the design standards for structural materials. The reliability indices in those earlier studies were determined for structural members based on a service period of 50 years. System reliabilities are higher to a degree that depends on structural redundancy and ductility. The probabilities represent, *in order of magnitude*, the associated annual member failure rates for those who would find this information useful in selecting a reliability target.

The load requirements in Sections 2.3.1–2.3.3 are supported by extensive peer-reviewed statistical databases, and the values of mean and coefficient of variation, μ_Q/Q_n and V_Q, are well established. This support may not exist for other loads that traditionally have not been covered by this standard. Similarly, the statistics used to determine μ_R/R_n and V_R should be consistent with the underlying material specification. When statistics are based on small-batch test programs, all reasonable sources of end-use variability should be incorporated in the sampling plan. The engineer should document the basis for all statistics selected in the analysis and submit the documentation for review by the Authority Having Jurisdiction. Such documents should be made part of the permanent design record.

The engineer is cautioned that load and resistance criteria necessary to achieve a reliability-based performance objective are coupled through the common term β in Eqs. (C2.3-1) and (C2.3-2). Adjustments to the load factors without corresponding adjustments to the resistance factors will lead to an unpredictable change in structural performance and reliability.

C2.3.6 Basic Combinations with Seismic Load Effects. The seismic load effect, E, is combined with the effects of other loads. For strength design, the load combinations in Section 2.3.6 with E include the horizontal and vertical seismic load effects of Sections 12.4.2.1 and 12.4.2.2, respectively. Similarly, the basic load combinations for allowable stress design in Section 2.4.8 with E include the same seismic load effects.

The seismic load effect including overstrength factor, E_m, is combined with other loads. The purpose for load combinations with overstrength factor is to approximate the maximum seismic load combination for the design of critical elements, including discontinuous systems, transfer beams and columns supporting discontinuous systems, and collectors. The allowable stress increase for load combinations with overstrength is to provide compatibility with past practice.

C2.4 LOAD COMBINATIONS FOR ALLOWABLE STRESS DESIGN

C2.4.1 Basic Combinations. The load combinations listed cover those loads for which specific values are given in other parts of this standard. Design should be based on the load combination causing the most unfavorable effect. In some cases, this may occur when one or more loads are not acting. No safety factors have been applied to these loads because such factors depend on the design philosophy adopted by the particular material specification. The principal load, or maximum variable load, in the load combinations is identified in Table C2.4-1.

Wind and earthquake loads need not be assumed to act simultaneously. However, the most unfavorable effects of each should be considered separately in design, where appropriate. In some instances, forces caused by wind might exceed those caused by earthquake, and ductility requirements might be determined by earthquake loads.

Load combinations (7) in Section 2.4.1 and (10) in Section 2.4.5 address the situation in which the effects of lateral or uplift forces counteract the effect of gravity loads. This action eliminates an inconsistency in the treatment of counteracting

Table C2.4-1. Principal Loads for Allowable Stress Design Load Combinations

Load Combination		Principal Load
1	D	D
2	$D + L$	L
3	$D + (L_r$ or S or $R)$	L_r or S or R
4	$D + 0.75L + 0.75(L_r$ or S or $R)$	L
5	$D + 0.6W$	W
6	$D + 0.75L + 0.75(0.6W) + 0.75(L_r$ or S or $R)$	W
7	$0.6D + 0.6W$	W
8	$D + 0.7E_v + 0.7E_{mh}$	E
9	$D + 0.525E_v + 0.525E_{mh} + 0.75L + 0.75S$	E
10	$0.6D - 0.7E_v + 0.7E_{mh}$	E

loads in allowable stress design and strength design and emphasizes the importance of checking stability. The reliability of structural components and systems in such a situation is determined mainly by the large variability in the destabilizing load (Ellingwood and Li 2009), and the factor 0.6 on dead load is necessary for maintaining comparable reliability between strength design and allowable stress design. The earthquake load effect is multiplied by 0.7 to align allowable stress design for earthquake effects with the definition of E in Section 11.3 of this standard, which is based on strength principles.

Most loads, other than dead loads, vary significantly with time. When these variable loads are combined with dead loads, their combined effect should be sufficient to reduce the risk of unsatisfactory performance to an acceptably low level. However, when more than one variable load is considered, it is extremely unlikely that they will all attain their maximum value at the same time (Turkstra and Madsen 1980). Accordingly, some reduction in the total of the combined load effects is appropriate. This reduction is accomplished through the 0.75 load combination factor. The 0.75 factor applies only to the variable loads because the dead loads (or stresses caused by dead loads) do not vary in time.

Some material design standards that permit a one-third increase in allowable stress for certain load combinations have justified that increase by this same concept. Where that is the case, simultaneous use of both the one-third increase in allowable stress and the 25% reduction in combined loads is unsafe and is not permitted. In contrast, allowable stress increases that are based upon duration of load or loading rate effects, which are independent concepts, may be combined with the reduction factor for combining multiple variable loads. In such cases, the increase is applied to the total stress, that is, the stress resulting from the combination of all loads.

In addition, certain material design standards permit a one-third increase in allowable stress for load combinations with one variable load where that variable is earthquake load. This standard handles allowable stress design for earthquake loads in a fashion to give results comparable to the strength design basis for earthquake loads as defined in Chapter 12 of this standard.

Exception (1) permits the companion load S appearing in combinations (4) and (6) to be the balanced snow load defined in Sections 7.3 for flat roofs and 7.4 for sloped roofs. Drifting and unbalanced snow loads, as principal loads, are covered by combination (3).

When wind forces act on a structure, the structural action causing uplift at the structure–foundation interface is less than would be computed from the peak lateral force because of area averaging. Area averaging of wind forces occurs for all structures. In the method used to determine the wind forces for enclosed structures, the area-averaging effect is already taken into account in the data analysis leading to the pressure coefficients C_p [or (GC_p)]. However, in the design of tanks and other industrial structures, the wind force coefficients, C_f, provided in the standard do not account for area averaging. For this reason, exception 2 permits the wind interface to be reduced by 10% in the design of nonbuilding structure foundations and to self-anchored ground-supported tanks. For different reasons, a similar approach is already provided for seismic actions in Section 2.4.1, Exception 2.

Exception 2 in Section 2.4.5 for special reinforced masonry walls is based upon the combination of three factors that yield a conservative design for overturning resistance under the seismic load combination:

1. The basic allowable stress for reinforcing steel is 40% of the specified yield.
2. The minimum reinforcement required in the vertical direction provides a protection against the circumstance where the dead and seismic loads result in a very small demand for tension reinforcement.
3. The maximum reinforcement limit prevents compression failure under overturning.

Of these, the low allowable stress in the reinforcing steel is the most significant. This exception should be deleted when and if the standard for design of masonry structures substantially increases the allowable stress in tension reinforcement.

C2.4.2 Load Combinations Including Flood Load. See Section C2.3.2. The multiplier on F_a aligns allowable stress design for flood load with strength design.

C2.4.3 Load Combinations Including Atmospheric Ice Loads. See Section C2.3.3.

C2.4.4 Load Combinations Including Self-Straining Forces and Effects. When using allowable stress design, determination of how self-straining forces and effects should be considered together with other loads should be based on the considerations discussed in Section C2.3.4. For typical situations, the following load combinations should be considered for evaluating the effects of self-straining forces and effects together with dead and live loads.

$$1.0D + 1.0T$$

$$1.0D + 0.75(L + T)$$

These combinations are not all-inclusive, and judgment is necessary in some situations. For example, where roof live loads or snow loads are significant and could conceivably occur simultaneously with self-straining forces, their effect should be included. The design should be based on the load combination causing the most unfavorable effect.

C2.5 LOAD COMBINATIONS FOR EXTRAORDINARY EVENTS

Section 2.5 advises the structural engineer that certain circumstances might require structures to be checked for low-probability events such as fire, explosions, and vehicular impact. Since the 1995 edition of ASCE Standard 7, Commentary C2.5 has provided a set of load combinations that were derived using a probabilistic basis similar to that used to develop the load combination requirements for ordinary loads in Section 2.3. In recent years, social and political events have led to an increasing desire on the part of architects, structural engineers, project developers, and regulatory authorities to enhance design and construction practices for certain buildings to provide additional structural robustness and to lessen the likelihood of disproportionate collapse if an abnormal event were to occur. Several federal, state, and local agencies have adopted policies that require new buildings and structures to be constructed with such enhancements of structural robustness (GSA 2003; DOD 2009). Robustness typically is assessed by notional removal of key load-bearing structural elements, followed by a structural analysis to assess the ability of the structure to bridge over the damage (often denoted alternative path analysis). Concurrently, advances in structural engineering for fire conditions (e.g., AISC 2010, Appendix 4) raise the prospect that new structural design requirements for fire safety will supplement the existing deemed-to-satisfy provisions in the next several years. To meet these needs, the load combinations for extraordinary events were moved in ASCE 7-10 (ASCE 2010) to Section 2.5 of this standard from Commentary C2.5, where they appeared in previous editions.

These provisions are not intended to supplant traditional approaches to ensure fire endurance based on standardized time–temperature curves and code-specified endurance times. Current code-specified endurance times are based on the ASTM E119 (2014) time–temperature curve under full allowable design load.

Extraordinary events arise from service or environmental conditions that traditionally are not considered explicitly in design of ordinary buildings and other structures. Such events are characterized by a low probability of occurrence and usually a short duration. Few buildings are ever exposed to such events, and statistical data to describe their magnitude and structural effects are rarely available. Included in the category of extraordinary events would be fire, explosions of volatile liquids or natural gas in building service systems, sabotage, vehicular impact, misuse by building occupants, subsidence (not settlement) of subsoil, and tornadoes. The occurrence of any of these events is likely to lead to structural damage or failure. If the structure is not properly designed and detailed, this local failure may initiate a chain reaction of failures that propagates throughout a major portion of the structure and leads to a potentially catastrophic partial or total collapse. Although all buildings are susceptible to such collapses in varying degrees, construction that lacks inherent continuity and ductility is particularly vulnerable (Taylor 1975; Breen and Siess 1979; Carper and Smilowitz 2006; Nair 2006; NIST 2007).

Good design practice requires that structures be robust and that their safety and performance not be sensitive to uncertainties in loads, environmental influences, and other situations not explicitly considered in design. The structural system should be designed in such a way that if an extraordinary event occurs, the probability of damage disproportionate to the original event is sufficiently small (Carper and Smilowitz 2006; NIST 2007). The philosophy of designing to limit the spread of damage rather than to prevent damage entirely is different from the traditional approach to designing to withstand dead, live, snow, and wind loads but is similar to the philosophy adopted in modern earthquake-resistant design.

In general, structural systems should be designed with sufficient continuity and ductility that alternate load paths can develop after individual member failure so that failure of the structure as a whole does not ensue. At a simple level, continuity can be achieved by requiring development of a minimum tie force, say 20 kN/m (1.37 kip/ft) between structural elements (NIST 2007). Member failures may be controlled by protective measures that ensure that no essential load-bearing member is made ineffective as a result of an accident, although this approach may be more difficult to implement. Where member failure would inevitably result in a disproportionate collapse, the member should be designed for a higher degree of reliability (NIST 2007).

Design limit states include loss of equilibrium as a rigid body, large deformations leading to significant second-order effects, yielding or rupture of members or connections, formation of a mechanism, and instability of members or the structure as a whole. These limit states are the same as those considered for other load events, but the load-resisting mechanisms in a damaged structure may be different, and sources of load-carrying capacity that normally would not be considered in ordinary ultimate limit states design, such as arch, membrane, or catenary action, may be included. The use of elastic analysis underestimates the load-carrying capacity of the structure (Marjanishvili and Agnew 2006). Materially or geometrically nonlinear or plastic analyses may be used, depending on the response of the structure to the actions.

Specific design provisions to control the effect of extraordinary loads and risk of progressive failure are developed with a probabilistic basis (Ellingwood and Leyendecker 1978; Ellingwood and Corotis 1991; Ellingwood and Dusenberry 2005). One can either reduce the likelihood of the extraordinary event or design the structure to withstand or absorb damage from the event if it occurs. Let F be the event of failure (damage or collapse) and A be the event that a structurally damaging event occurs. The probability of failure due to event A is

$$P_f = P(F|A)P(A) \qquad \text{(C2.5-1)}$$

in which $P(F|A)$ is the conditional probability of failure of a damaged structure and $P(A)$ is the probability of occurrence of event A. The separation of $P(F|A)$ and $P(A)$ allows one to focus on strategies for reducing risk. $P(A)$ depends on siting, controlling the use of hazardous substances, limiting access, and other actions that are essentially independent of structural design. In contrast, $P(F|A)$ depends on structural design measures ranging from minimum provisions for continuity to a complete post-damage structural evaluation.

The probability, $P(A)$, depends on the specific hazard. Limited data for severe fires, gas explosions, bomb explosions, and vehicular collisions indicate that the event probability depends on building size, measured in dwelling units or square footage, and ranges from about 0.2×10^{-6}/dwelling unit/year to about 8.0×10^{-6}/dwelling unit/year (NIST 2007). Thus, the probability that a building structure is affected may depend on the number of dwelling units (or square footage) in the building. If one were to set the conditional limit state probability, $P(F|A) = 0.05$–0.10, however, the annual probability of structural failure from Eq. (C2.5-1) would be less than 10^{-6}, placing the risk in the low-magnitude background along with risks from rare accidents (Pate-Cornell 1994).

Design requirements corresponding to this desired $P(F|A)$ can be developed using first-order reliability analysis if the limit state function describing structural behavior is available (Ellingwood and Dusenberry 2005). The structural action (force or constrained deformation) resulting from extraordinary event A used in design is denoted A_k. Only limited data are available to define the frequency distribution of the load (NIST 2007; Ellingwood and Dusenberry 2005). The uncertainty in the load caused by the extraordinary event is encompassed in the selection of a conservative A_k, and thus the load factor on A_k is set equal to 1.0, as is done in the earthquake load combinations in Section 2.3. The dead load is multiplied by the factor 0.9 if it has a stabilizing effect; otherwise, the load factor is 1.2, as it is with the ordinary combinations in Sections 2.3.1 and 2.3.6. Load factors less than 1.0 on the companion actions reflect the small probability of a joint occurrence of the extraordinary load and the design live, snow, or wind load. The companion actions $0.5L$ and $0.2S$ correspond, approximately, to the mean of the yearly maximum live and snow loads (Chalk and Corotis 1980; Ellingwood 1981). The companion action in Eq. (2.5-1) includes only snow load because the probability of a coincidence of A_k with L_r or R, which have short durations in comparison with S, is negligible. A similar set of load combinations for extraordinary events appears in *Eurocode 1* (2006).

The term $0.2W$ that previously appeared in these combinations has been removed and has been replaced by a requirement to check lateral stability. One approach for meeting this requirement, which is based on recommendations of the Structural Stability Research Council (Galambos 1998), is to apply lateral notional forces, $N_i = 0.002\Sigma P_i$, at level i, in which ΣP_i = gravity force from Eq. (2.5-1) or (2.5-2) act at level i, in combination with the loads stipulated in Eq. (2.5-1) or (2.5-2). Note that Eq. (1.4-1) stipulates that when checking general structural integrity, the lateral forces acting on an intact structure shall equal $0.01W_x$, where W_x is the dead load at level x.

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